Understanding the Blockchain Interoperability Graph based on Cryptocurrency Price Correlation

Ori Mazor and Ori Rottenstreich

***Abstract*—Cryptocurrencies have gained high popularity in recent years, with over 9000 of them, including major ones such as Bitcoin and Ether. Each cryptocurrency is implemented on one blockchain or over several such networks. Recently, various technologies known as blockchain interoperability have been developed to connect these different blockchains and create an interconnected blockchain ecosystem. This paper aims to provide insights on the blockchain ecosystem and the connection between blockchains that we refer to as the interoperability graph. Our approach is based on the analysis of the correlation between cryptocurrencies implemented over the different blockchains. We examine over 4800 cryptocurrencies implemented on 76 blockchains and their daily prices over a year. This experimental study has potential implications for decentralized finance (DeFi), including portfolio investment strategies and risk management.**

***Index Terms*—Blockchain, Interoperability, Graph analysis.**

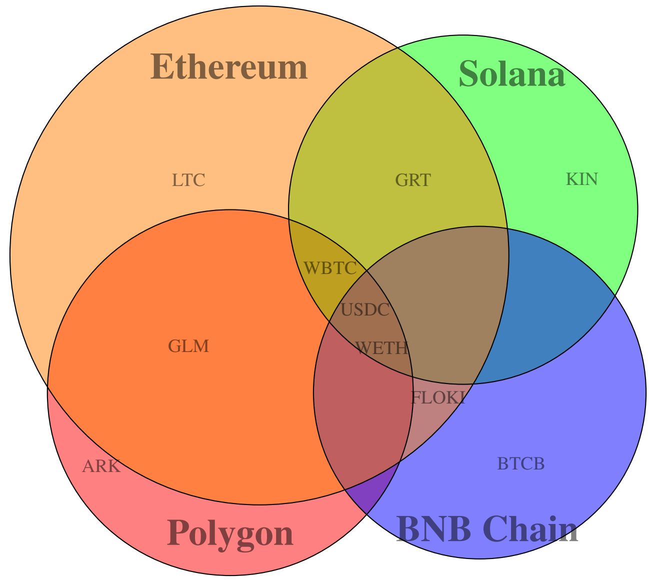


Fig. 1. Venn diagram of four major blockchains Ethereum, Polygon, Binance Smart Chain (BNB Chain) and Solana with their implemented cryptocurrencies. The cryptocurrencies referred to are Wrapped Bitcoin (WBTC), USD Coin (USDC), Litecoin (LTC), The Graph (GRT), Kin (KIN), Floki Inu (FLOKI), Bitcoin BEP2 (BTCB), Wrapped Ether (WETH), Golem (GLM), and Ark (ARK).

I. Introduction

Cryptocurrencies, with a market capitalization of approximately 4.1 trillion USD as of August 2025 [1], have a high impact on the global economy and offer new investment opportunities [2], [3]. The development of this sector is mainly due to the popularity of Bitcoin [4], which was established in 2008 with the introduction of blockchain technology. Since then, numerous blockchains have been introduced, serving various functions and supporting their unique cryptocurrencies, such as Ethereum, Polygon, BNB Chain, and Solana. Ethereum [5], recognized as a significant blockchain innovation, introduced smart contracts. These programs operate on the blockchain, consistently maintaining their state during executions.

Smart contracts enable the exchange of cryptocurrencies across various blockchains and their respective implementations on these networks. Numerous mechanisms, typically known as interoperability or cross-chain technologies, have been developed and examined to facilitate inter-blockchain communication [6]–[11]. Blockchain interoperability allows blockchain systems to exchange data, messages, and digital assets with mitigated security risks. This advancement is considered critical for the wide adoption of blockchain technology as it has transformed the blockchain ecosystem from isolated data islands to an interconnected network.

Our study is motivated by the importance and emergence of blockchain interoperability. We aim to provide insights into the interoperability between major blockchains and the implied blockchain ecosystem as a whole based on such connections. We view the blockchain ecosystem as the *Blockchain Interoperability Graph*. In the graph, nodes represent different blockchains, while edges represent the level of interoperability between a pair of blockchains. We aim to offer insights into the graph by analyzing the correlations between cryptocurrencies implemented across various blockchains. Our view of the blockchain interoperability focuses only on correlation of cryptocurrency prices and, accrodingly, does not aim to describe the interoperability technological solutions that connect two blockchains. Unlike traditional studies on pairs of cryptocurrencies our analysis focuses on blockchains, based on cryptocurrencies associated with them.

Contributions. We present the following contributions.

- We overview the blockchain ecosystem with terminology and main players (Section II).
- We present a methodology for measuring interoperability between blockchains based on cryptocurrency prices (Section III).
- We apply the methodology to analyze the blockchain ecosystem in terms of its interoperability (Section IV).
- We enhance the interoperability analysis while focusing on major blockchains (Section V).
- We discuss investment applications of this study (Section VI).

Ori Mazor and Ori Rottenstreich are with the Technion, Haifa, Israel. (Emails: orimazor@campus.technion.ac.il, or@technion.ac.il).

II. Background - The Blockchain Ecosystem

In this Section, we overview common terminology in the area of blockchain interoperability [12], describe algorithmic methods we use for analysis, and detail several key blockchains in the blockchain ecosystem.

A. Terminology of blockchain interoperability and methods

Cross-chain DEXs are decentralized exchanges that enable the exchange of assets between different blockchains. Cross-chain DEXs often rely on tools known as cross-chain bridges for their implementation. These bridges can be centralized or decentralized based on the protocol implementation.

Wrapped Tokens are typically cryptocurrencies that are pegged to the value of another original cryptocurrency or asset, designed for interoperability with different blockchains.

Cryptocurrencies can have different implementations on several blockchains, which provides compatibility with cross-chain technologies and enables further utilization of cryptocurrencies on each distinct blockchain. Figure 1 displays various cryptocurrencies implemented across multiple blockchains as detailed on CoinMarketCap [13]. Throughout the paper, we categorize the cryptocurrencies by their parent blockchain based on CoinMarketCap API [14].

Layers 0-2: Layer 0, Layer 1, and Layer 2 are hierarchical concepts that articulate the architecture and operation of the blockchains.

Layer 0 blockchains serve as infrastructure for other blockchains, enabling them to communicate and scale, for example, Polkadot and Cosmos.

Layer 1 represents the main blockchain layer, responsible for the key functionalities of the blockchain, such as consensus. Primary Layer 1 blockchains, for example, include Bitcoin and Ethereum.

Layer 2 blockchains are built on top of Layer 1 primarily to enhance scalability. Optimism is an example of a Layer 2 blockchain.

Sidechains are secondary blockchains compatible with the main blockchain network. They facilitate communication with a main blockchain using a two-way peg. A two-way peg is a mechanism that facilitates the two-directional movement of assets between the main chain and the sidechain, according to a set or pre-established exchange rate [15] such as Polygon.

We use several simple mathematical concepts for the analysis. These concepts are widely used to determine price trends across different financial fields.

Log Return: is a way to measure the relative change in the price of an asset over a time period.

Pearson's Correlation Coefficient: A measurement of the linear correlation between two quantitative variables [16].

Minimum Spanning Tree (MST): A spanning tree is a subgraph where all vertices are connected without cycles, using the minimum number of edges required to create it. A minimum spanning tree is a spanning tree with the minimal total weight among all possible spanning trees [17], [18].

B. Key blockchains in the blockchain ecosystem

Arbitrum: is a Layer 2 blockchain for Ethereum focuses primarily on providing scalability to Ethereum.

BNB Chain: is a blockchain developed by Binance, one of the world's largest cryptocurrency exchanges, designed with a focus on decentralized finance (DeFi) [19], [20].

Cosmos: is a Layer 0 blockchain, enabling various blockchains to communicate and scale effectively [21].

Ethereum Classic: is a blockchain that originated as a fork of the Ethereum network following the 2016 DAO hack.

Harmony: is a blockchain designed to support decentralized applications (DApps) with a focus on scalability, security, and energy efficiency, utilizing a unique sharding architecture [22].

NEAR: is a blockchain designed to be a fast and secure, focusing on scalability and user experience. [23].

Optimism: is a Layer 2 blockchain for Ethereum that uses optimistic rollups to enable fast and cost-effective transactions.

Polkadot: is a layer 0 blockchain designed to connect a multitude of blockchains, facilitating their seamless integration and collective operation on a large scale [24].

Solana: is a blockchain designed for scalable and fast applications. It uses a consensus protocol that combines Proof of History (PoH) with Proof of Stake (PoS) to achieve high throughput and low latency [25], [26].

Theta Network: The Theta Network's blockchain is designed primarily to address the needs of the media and entertainment industry [27].

Tron: is a blockchain focused on creating a decentralized internet with decentralized applications (DApps) [28].

VeChain: is a blockchain platform focused on supply chain [29].

III. A Methodology for Measuring Interoperability based on Cryptocurrency Prices

Methodology Overview. The primary objective of this methodology is to develop an algorithmic and mathematical framework for analyzing the blockchain ecosystem through price correlations among blockchains. We represent the ecosystem as a graph, where the vertices denote blockchains, and the edges are weighted using a mathematical measure of price correlations between cryptocurrencies. We propose algorithms to construct the cryptocurrency correlation graph, where vertices represent cryptocurrencies and edge weights correspond to price correlations between them. This graph is then utilized to derive the blockchain correlation graph. Furthermore, we introduce methods to analyze both graphs to derive meaningful insights into their structure and dynamics.

Detailed Description of the Methodology. We model the blockchain ecosystem as the *blockchain interoperability graph*, an (undirected) graph in which nodes represent blockchains. In such a graph, an edge exists between two blockchains upon the existence of an interoperability mechanism between the pair of blockchains that allows the efficient exchange of data and digital assets. The graph is weighted such that an edge weight is small if the interoperability mechanism is time efficient, has low cost and low-security risks. In such an ecosystem, two blockchains that are not directly connected can communicate indirectly through a third blockchain or a path of them, often with higher costs in terms of delay, risk, etc.

Towards understanding the blockchain interoperability graph, we gathered the daily closing prices of 4828 cryptocurrencies from 76 different blockchains for approximately a year, from January 21, 2023, to January 17, 2024, from CoinMarketCap [13]. We aim to provide insights on the blockchain ecosystem by studying the following:

- The correlation between cryptocurrencies implemented on the same parent blockchain and the correlation between cryptocurrencies implemented on different parent blockchains.
- The correlation between a pair of blockchains by the correlation of cryptocurrencies for which these blockchains serve as the parent blockchain.

To study the correlation between cryptocurrencies, we use logarithmic return (log return) and Pearson's correlation coefficient as it is widely used [30], [31]. For cryptocurrency i, the closing price on day t is defined as $p_i(t)$. The daily log return, at day t for cryptocurrency i is therefore:

$$R_i(t) = ln\left(\frac{p_i(t)}{p_i(t-1)}\right) = ln(p_i(t)) - ln(p_i(t-1))$$

For a specific timeframe T, R_i represents a vector of log returns. The correlation coefficient (Pearson's correlation coefficient) that measures the correlation between cryptocurrencies i and j over timeframe T is calculated by cross-correlation function. We denote $\sigma_i = \sqrt{E(R_i^2) - E(R_i)^2}$ the standard deviation and $E(\cdot)$ the expected value. We refer to the cross-correlation function as

$$C_{i,j} = \frac{E(R_i R_j) - E(R_i) \cdot E(R_j)}{\sigma_i \cdot \sigma_j}.$$

Note that C is symmetric as $C_{i,j} = C_{j,i}$ and $C_{i,j} \in [-1, 1]$ where -1 indicates maximum anti-correlation, 0 absence of correlation and 1 maximum correlation. We use a metric for Pearson's correlation value based on Evans [32] to measure the correlation strength, as summarized in Table I. We denote the parent blockchain of cryptocurrency i as B_i, and τ as the threshold. We define:

$$C^\tau_{i,j} = \begin{cases} C_{i,j}, & \text{if } |C_{i,j}| \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

If the correlation is not above the threshold, we consider the cryptocurrencies uncorrelated. The distance between cryptocurrencies i and j is defined as follows:

$$L^\tau_{i,j} = 1 - (C^\tau_{i,i})^2,$$

where $L^\tau_{i,j} \in [0, 1]$ as chosen in [33]. This transformation satisfies an intuitive property where strongly correlated cryptocurrencies receive low distance values. For two cryptocurrencies i and j implemented on some shared blockchain B_k, we denote

$$L^{\tau,k}_{i,j} = L^\tau_{i,j}.$$

To represent the distance matrix of the cryptocurrencies within the same parent blockchain.

TABLE I
Evan's metric for measuring the strength of Pearson's correlation

Correlation Strength	Correlation	Anti-correlation
Very weak	0.00 - 0.19	-0.19 - 0.00
Weak	0.20 - 0.39	-0.39 - 0.20
Moderate	0.40 - 0.59	-0.59 - 0.40
Strong	0.60 - 0.79	-0.79 - 0.60
Very strong	0.80 - 1.00	-1.00 - 0.80

To study the correlation between cryptocurrencies, we create an undirected graph in which each cryptocurrency is represented by a node, and the edges indicate the correlation value between the cryptocurrencies. The algorithm for building the graph is described in Algorithm 1. The algorithm receives start date and end date and returns the graph. *cryptoLogReturnVector* is a dictionary where the keys represent the cryptocurrencies, and the values are the corresponding vectors of log returns. *goneByCryptos* are cryptocurrencies whose correlation to the other cryptocurrencies was calculated. *PricesAllCryptos* receives a start date and an end date and returns a dictionary where the keys are the cryptocurrencies, and the values are dictionaries of prices by date. *CalcCorrelation* calculates the Pearson's correlation coefficient based on the cryptocurrencies and the log return vector.

Algorithm 1: Build correlation graph of cryptocurrencies ($startDate, endDate$)

```
Init an empty graph G
Init empty dictionary cryptoLogReturnVector
Init empty goneByCryptos
pricesCryptos = PricesAllCryptos(startDate, endDate)
cryptos = pricesCryptos.keys()
for c ∈ cryptos do
    Add c to goneByCryptos
    if c ∉ cryptoLogReturnVector then
        cryptoLogReturnVector[c] ←
          CalcReturn(startDate, endDate, pricesCryptos, c)
    for c′ ∉ {cryptos \ goneByCryptos} do
        if c′ ∉ cryptoLogReturnVector then
            cryptoLogReturnVector[c′] ←
              CalcReturn(startDate, endDate, pricesCryptos, c′)
        e ← CalcCorrelation(c, c′, cryptoLogReturnVector)
        Add e to G
return G
```

Algorithm 2 receives start date, end date, dictionary of prices per cryptocurrency, and the cryptocurrency. It returns the vector of log return values of the cryptocurrency in the timeframe.

Algorithm 2: $CalcReturn(startDate, endDate, prices, c)$

```
Init empty list returnsList
for from t = startDate + 1 to endDate do
    returnValue ← ln(prices[c][t]) − ln(prices[c][t − 1])
    Add returnValue to returnsList
return returnsList
```

For each cryptocurrency we divide all other cryptocurrencies into two groups:

Group 1: Cryptocurrencies that share the same parent blockchain as the cryptocurrency.

Group 2: Cryptocurrencies with a different parent blockchain as of the cryptocurrency.

An edge that connects a cryptocurrency with another cryptocurrency from group 1 is considered an inner edge, as it connects two cryptocurrencies in the same parent blockchain. An edge that connects a cryptocurrency with another cryptocurrency from group 2 is considered an intersecting edge as it connects two cryptocurrencies implemented on different parent blockchains.

We normalize each cryptocurrency's inner edges correlation values to those of its intersecting edges correlation values. We get the normalized value for the inner edges as *InDominance* and the intersecting edges as *OutDominance* (measured in percentage as a function of τ) as follows:

$$\begin{aligned} In &= \{(i,j) | j \in B_i, j \neq i\} \\ Out &= \{(i,j) | j \notin B_i\} \\ InHeavy(\tau) &= \{(i,j) | j \in B_i, j \neq i, |C_{i,j}| \geq \tau\} \\ OutHeavy(\tau) &= \{(i,j) | j \notin B_i, |C_{i,j}| \geq \tau\} \\ InHeavyRatio(\tau) &= \frac{|InHeavy(\tau)|}{|In|} \\ OutHeavyRatio(\tau) &= \frac{|OutHeavy(\tau)|}{|Out|} \end{aligned}$$

The following two normalized correlation values have percentage as their unit.

$$\begin{aligned} InDominance(\tau) &= \frac{100 \cdot InHeavyRatio(\tau)}{InHeavyRatio(\tau) + OutHeavyRatio(\tau)} \\ OutDominance(\tau) &= 100 - InDominance(\tau) \end{aligned}$$

We use these measurements in Section IV to provide insights into the relationship between cryptocurrencies from group 1 and group 2. We measure the correlation between each pair of blockchains based on the correlation of cryptocurrency prices for which these blockchains serve as the parent blockchain. We define $S^{\tau}_{k,m}$ the pairs of cryptocurrencies with a correlation value greater than or equal to τ between blockchains k and m as follows:

$$S^{\tau}_{k,m} = \{(i,j) | i \in B_k, j \in B_m, |C_{i,j}| \geq \tau\}$$

We use $S^{\tau}_{k,m}$ to define the correlation between blockchain k and blockchain m for threshold τ:

$$N^{\tau}_{k,m} = \frac{1}{|S^{\tau}_{k,m}|} \cdot \sum_{(i,j) \in S^{\tau}_{k,m}} C_{i,j}$$

If there are no cryptocurrencies correlated between the blockchains, meaning $S^{\tau}_{k,m} = \emptyset$, then $N^{\tau}_{k,m} = 0$, as we consider the blockchains to be uncorrelated.

To study the blockchain behavior, we use the Minimum Spanning Tree (MST), which effectively reduces the dimensionality of the correlation matrix by retaining only $N-1$ connections out of $\frac{N(N-1)}{2}$ possible by considering the significance of the connections while keeping connectivity among all nodes. This approach acts as a noise filter, preserving meaningful relationships while discarding spurious correlations. The MST also reveals hierarchical structures that often correspond to economic sectors [34]. Its graphical representation facilitates intuitive understanding of asset interdependencies [35], and by applying MST over rolling time windows, one can monitor the dynamic evolution of market structures and systemic risk [36]. The method has been widely validated in the literature as an effective tool for financial network analysis [37]–[39]. To employ the MST using the blockchain correlation, we first transform the blockchain correlation matrix N into a distance correlation matrix. We apply the following transformation to create a distance matrix. The distance between blockchains k and m is defined as follows:

$$D^{\tau}_{k,m} = 1 - (N^{\tau}_{k,m})^2$$

where $D^{\tau}_{k,m} \in [0,1]$ as chosen in [33]. This transformation satisfies an intuitive property where strongly correlated blockchains receive low distance values.

We try to understand from D the blockchains collective behavior and community structure. We use the Clauset-Newman-Moore [40] community detection algorithm to detect the distinct communities within the MST as shown in Section IV. These communities provide insights into the interoperability graph as they represent groups of correlated blockchains, which are influenced by communication between these blockchains. We use the same methods with L on cryptocurrencies whose parent blockchain is Ethereum or BNB. We apply Kruskal's algorithm to compute the minimum spanning tree (MST) on L and the Clauset-Newman-Moore algorithm on the Minimum Spanning Tree (MST) created for the Ethereum and BNB blockchains to gain deeper insights into the correlation behavior within each blockchain. We measure the centrality of a graph based on [41]. In a graph $G = (V, E)$, where $|V|$ represents the number of vertices and $|E|$ represents the number of edges, the degree centrality $C_D(v)$ of a vertex v is defined by its degree, $C_D(v) = \deg(v)$. The vertex with the highest degree centrality in G is denoted as v^*. The overall centrality of the graph is quantified by the formula:

$$C_D(G) = \frac{\sum_{i=1}^{|V|} [C_D(v^*) - C_D(v_i)]}{|V|^2 - 3|V| + 2}$$

This equation calculates the degree centralization of G, comparing the highest degree centrality to that of all other vertices, normalized over a function of the total number of vertices. We denote G' as the subgraph of a community in the MST. We consider communities that are centralized with $C_D(G') \geq 0.5$.

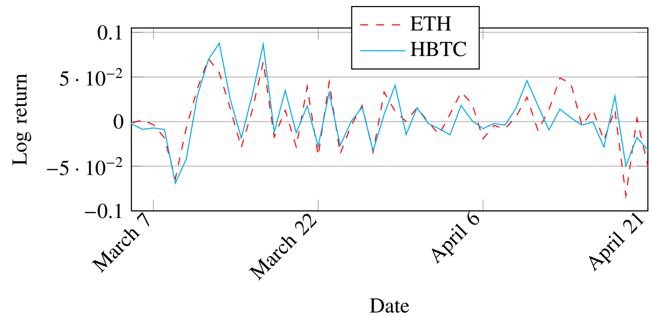
IV. Data Analysis of the Blockchain Ecosystem

In Table II, we present the 76 blockchains by their number of analyzed cryptocurrencies, and their layer. Layer 0 blockchains function as infrastructure for other blockchains implemented on top of them.

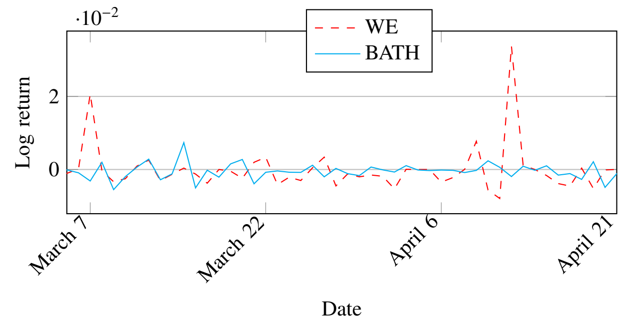
Using correlation data from January 21, 2023, to January 17, 2024, we construct the correlation graph presented in Section III which serves as the basis for our analysis. We divide this section into two subsections: In subsection IV-A, we demonstrate that cryptocurrencies tend to be more correlated within their parent blockchain. In subsection IV-B, we

TABLE II
BLOCKCHAINS STATISTICS (A TOTAL OF 76 BLOCKCHAINS).

Name	# Coins analyzed	Layer
Bitcoin	1	1
Ethereum	2303	1
BNB	1430	1
Solana	170	1
XRP Ledger	14	1
Cardano	24	1
Avalanche C-Chain	95	-
Dogechain	5	-
Polkadot	2	0
Tron	52	1
Ton	4	1
Polygon	187	2
Bitcoin Cash	2	1
Cosmos	2	0
ICP	2	-
Near	9	1
Stellar	11	1
Injective	1	2
Hedera	4	1
Ethereum Classic	2	1
Aptos	11	1
VeChain	3	1
Cronos	30	1
Optimism	9	2
Stacks	2	-
Elrond	19	1
Algorand	14	1
Fantom	54	1
Theta Network	2	1
Neo	8	1
EOS	8	1
Tezos	11	1
Klaytn	48	1
Osmosis	16	1
XDC Network	9	1
Conflux	2	1
IoTeX	5	1
Astar	2	1
Zilliqa	8	1
Nem	2	-
Celo	15	1
Waves	8	-
Moonbeam	9	1
Arbitrum	27	2
Harmony	14	-
Ont	2	-
RSK RBTC	3	2
Secret	2	1
Elastos	2	-
Everscale	5	1
Viction	2	1
Heco	19	-
Moonriver	6	1
Fusion network	2	1
Wanchain	4	1
Telos	3	1
Kardiachain	3	-
Chiliz	44	1
Bitcichain	22	-
Arbitrum Nova	1	2
Terra Classic	4	1
Songbird Network	2	1
Althash	2	-
Canto	3	1
OKExChain	5	1
Avalanche DFK	3	-
Aurora	2	2
Gnosis Chain	5	1
Step	2	-
Metis Andromeda	9	2
Fuse	2	1
Oasis Network	3	1
KCC	4	-
Sora	3	-
Bitgert	2	1
zkSync Era	1	2



(a) Log returns of ETH and HBTC over time yield a correlation of 0.8



(b) Log returns of WE and BATH over time yield a correlation of -0.8

Fig. 2. Log returns of four cryptocurrencies: ETH, HBTC, BATH, and WE by their correlation

study the dynamics between blockchains and the interoperability graph through the correlation of the cryptocurrencies implemented on them.

A. Cryptocurrencies correlations

In this subsection, we study the correlation between cryptocurrencies in terms of their parent blockchain implementation. We first demonstrate four different cryptocurrencies and their log returns. A positive correlation indicates the same log return behavior, while a negative one indicates anti-correlation behavior, meaning the log return behavior is not aligned. We demonstrate very strong and very weak correlations (based on Evans measure) of cryptocurrencies, which we calculate their correlation over the analyzed timeframe (from January 21, 2023, to January 17, 2024). We show their log returns values from March 21, 2023, to April 17, 2023, in Figure 2.

WE functions as a utility token within the WeBuy platform, which focuses on NFT-based commerce and rentals, while BATH is the native token of Battle Hero, a blockchain-based play-to-earn game. A high correlation is observed between Ether (ETH) and Huobi BTC (HBTC), while anti-correlation exists between WE and BATH. This may be the case due to several reasons. ETH serves as the native token of the Ethereum blockchain, whereas HBTC is a wrapped version of Bitcoin on Ethereum, backed 1:1 by BTC. Given that HBTC is designed to mirror the price of BTC, and that BTC and ETH frequently respond to similar macroeconomic factors and investor sentiment, their correlated behavior may be attributed to these shared influences. Additionally, their concurrent involvement in decentralized finance (DeFi) protocols may further reinforce this relationship. In contrast, WE and BATH

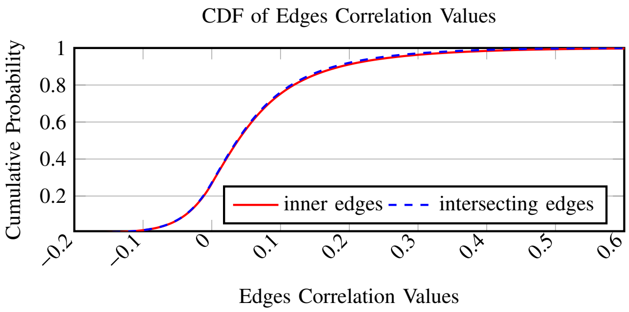


Fig. 3. CDF of inner and intersecting edges correlation values

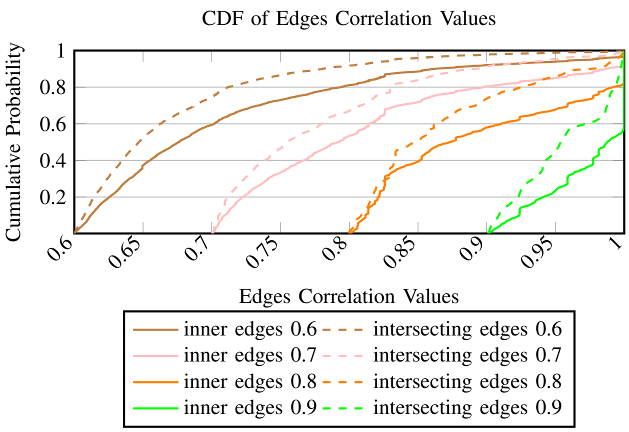


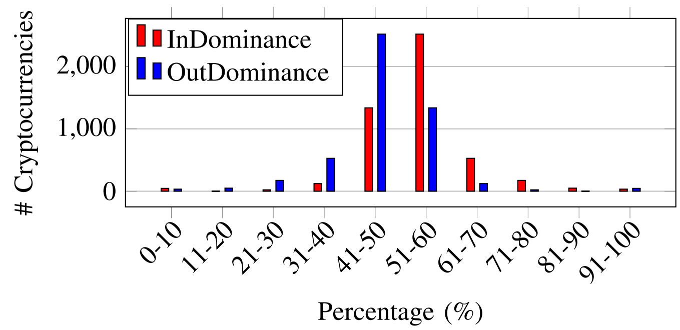
Fig. 4. CDF by threshold of inner and intersecting edges correlation values

appear to exhibit opposing price trends. These assets serve distinct purposes and target different user communities namely NFT commerce versus blockchain gaming. It is therefore possible that capital rotation, shifting market narratives, or divergent adoption trends contribute to their inverse behavior. Further research is needed to better understand the underlying mechanisms.

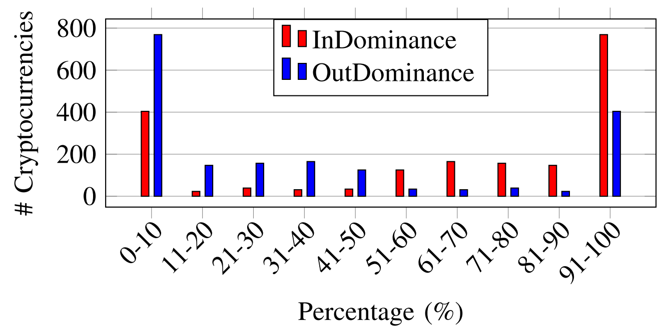
We focus on cryptocurrencies with strong and moderate correlations by applying various correlation thresholds. This approach helps us focus on the most relevant cryptocurrencies, as uncorrelated ones often do not provide valuable insights into the underlying dynamics of cryptocurrency correlations.

In Figure 3 and Figure 4 we show the cumulative distribution function (CDF) of the correlation values of intersecting edges and inner edges per threshold. In Figure 3, we show that most analyzed cryptocurrencies are not correlated as most have correlation values between -0.1 and 0.2. In Figure 4, we show that inner edges have higher correlation values in all the strong thresholds, indicating that cryptocurrencies tend to have higher correlation values with cryptocurrencies implemented on the same parent blockchain. We observe a spike around the correlation of 1, we assume it reflects structural relationships—such as wrapped tokens, synthetic assets, pegged stablecoins, or other instruments designed to mirror or represent another asset's price. These links naturally produce perfect or near-perfect correlations.

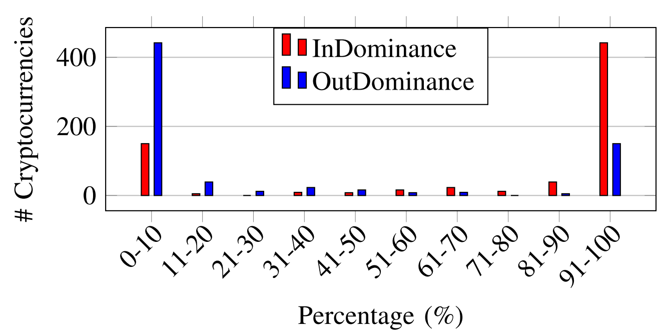
In Figure 5, we present for various values of the threshold τ, the distribution (over the analyzed cryptocurrencies) of the values of *InDominance*(τ) and *OutDominance*(τ), as defined in Section III. We aim to provide insights into the correlations of cryptocurrencies both within their parent blockchain and outside of it. Intuitively, the relationship between threshold values and the two dominance values provides insights on whether belonging to the same parent blockchain impacts the



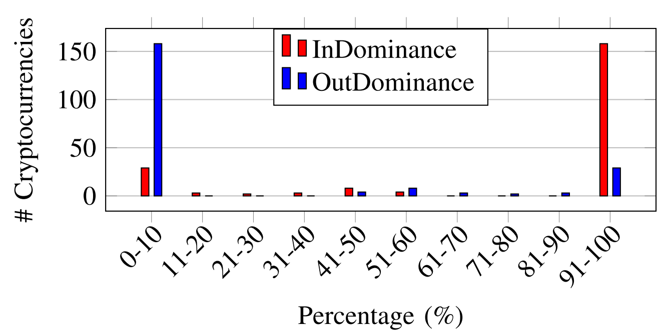
(a) Normalized edges correlation values for threshold $\tau = 0.1$



(b) Normalized edges correlation values for threshold $\tau = 0.5$

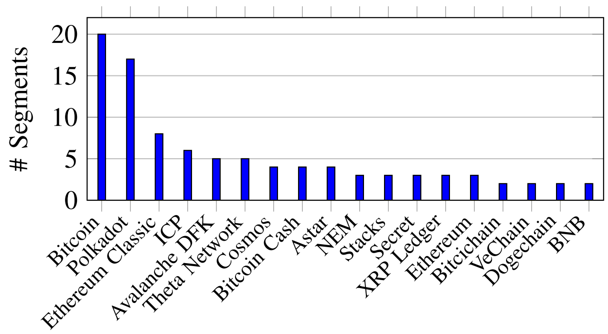


(c) Normalized edges correlation values for threshold $\tau = 0.7$

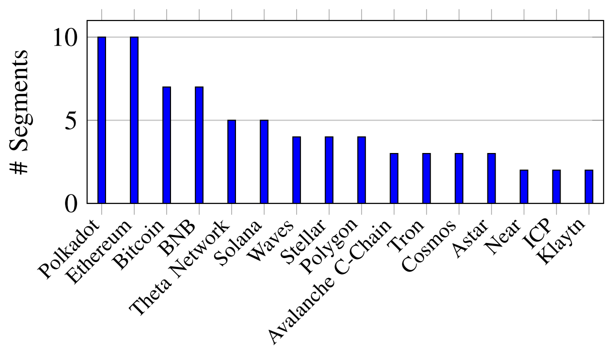


(d) Normalized edges correlation values for threshold $\tau = 0.9$

Fig. 5. Distribution of normalized edges correlation values for various threshold values



(a) Blockchain community centers with $\tau = 0.4$.



(b) Blockchain community centers with $\tau = 0.6$.

Fig. 6. Comparison of blockchain community centers at different thresholds of τ.

correlation of cryptocurrency prices. Recall that the sum of the two dominance values equals 100.

In Figure 5(a), we show that most cryptocurrencies with a correlation greater than 0.1 tend to have the same correlation to cryptocurrencies implemented on their parent blockchain and cryptocurrencies implemented on other blockchains. This is reasonable as most cryptocurrencies are not correlated to each other, as we showed in Figure 3; therefore, they do not tend to be correlated to either. For strongly correlated cryptocurrency, we conclude the following from Figure 5(c) and Figure 5(d):

- Cryptocurrencies tend to be more correlated within their parent blockchain as their *InDominance* value tends to be higher.
- Most cryptocurrencies with strong correlation tend to be either strongly correlated to the cryptocurrencies implemented on their parent blockchain or cryptocurrencies on other blockchains as most cryptocurrencies' *InDominance* and *OutDominance* values are in the bins $91-100$ or in bins $0-10$.

Overall, cryptocurrencies tend to be highly correlated with cryptocurrencies within their parent blockchain and tend to be either highly correlated to cryptocurrencies implemented on their parent blockchain or to cryptocurrencies implemented on other blockchains.

B. Analysis of the MSTs of the blockchain ecosystem

In this subsection, we study the blockchain ecosystem based on the correlation between blockchains as formulated in Section III, with thresholds values of $\tau = 0.4$ and $\tau = 0.6$ (moderate correlation and strong correlation). We represent the blockchain ecosystem as a graph where a node is a blockchain and an edge represents the correlation between the blockchains. We use the Kruskal and Clauset-Newman-Moore community detection algorithm to study the communities.

We analyze the structure of communities across several time frames within a year, spanning from January 21, 2023, to January 17, 2024. To observe the effects of varying durations on key properties of these communities, we segmented the year into the following periods: 31 days, 60 days, 90 days, 181 days, and 362 days we denote them as P. For each period, we refer to complete and disjoint segments that can be found in the total of 362 days that together contribute $\lfloor 362/31 \rfloor + \lfloor 362/60 \rfloor + \lfloor 362/90 \rfloor + \lfloor 362/181 \rfloor + \lfloor 362/362 \rfloor = 11 + 6 + 4 + 2 + 1 = 24$ segments.

We focus in our study on the following key properties:

1) Centers of communities
2) Consistency of communities over time
3) Consistency of correlation over time
4) Overall structure of communities

This approach explains how different timeframes influence essential aspects of community dynamics and structure.

Centers of Communities. We measure the community centers as described in Section III for each segment in P. We count the number of segments in which each blockchain is the center of a community. Our results are shown in Figure 6. In Figures 6(a) and 6(b), Bitcoin and Polkadot emerge as leading blockchains at the core of the centralized communities they induce. As shown in Figure 6(b), increasing the threshold τ causes Ethereum and BNB to become among the top centralized blockchains. This shift is expected given the extensive range of cryptocurrencies associated with these blockchains. Including only the more correlated ones from each blockchain supports their significant interconnections with a larger number of blockchains.

Consistency of Communities Over Time. The number of communities varies over time, typically ranging from 7 to 10. We assess the similarity between communities across adjacent periods for each category in P. The similarity of each community is defined as the highest Jaccard similarity value observed with any community in the following period. For example, if at timeframe t we have a community A composed of the blockchains $\{a, b\}$ and at timeframe t' we have a community B composed of the blockchains $\{a, b, d\}$, we calculate their similarity as follows:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|\{a, b\}|}{|\{a, b, d\}|} = 66.67\%.$$

On average, the similarity between communities is primarily between 18% and 20%, indicating that despite variations across different timeframes, there remains a consistent level of similarity that can be leveraged for insightful analyses. We also observe a consistency in the centers of communities, which generally ranges between 17% and 20%.

Consistency of Correlation Over Time. We assess the consistency of the MSTs over time by identifying edges that recur in adjacent periods for each category in P. The results show a similarity of primarily between 10% and 20%, indicating that approximately between 10% and 20% of the edges in two MSTs are the same, although with different correlation values. This suggests that while the structure of the blockchain ecosystem, as represented by MSTs, is volatile, it still retains key properties, such as the centers of communities.

Overall Structure of Communities. We now further analyze three timeframes of MSTs, as they span over extended periods: from January 21, 2023, to January 17, 2024 (as shown in Figure 14 in the Appendix), from January 21, 2023, to July 18, 2023 (in Figure 15 also in the Appendix), and from July 19, 2023, to January 17, 2024 (in Figure 16 in the Appendix as well). We observe ten communities in Figure 14, six of which have a central blockchain: Bitcoin, BNB, Harmony, Polkadot, VeChain, and Ethereum. The structure of the communities varies across different timeframes. However, there are key properties that we observe throughout these timeframes.

Bitcoin, Ethereum, and Polkadot are at the center of the communities they induce. This is reasonable because Bitcoin, as the inaugural cryptocurrency, continues to play a pivotal role in influencing the pricing trends across the cryptocurrency market. TVL is a metric that measures the cumulative USD value of digital assets locked or staked on a blockchain via decentralized finance (DeFi) platforms or decentralized applications (dApps). Blockchains with a higher TVL are considered more valuable and secure [42]. TVL values are based on data from DefiLlama [43] in July 2024. As TVL values might be hard to compute (e.g., due to the potential double count of assets), we just rely on how they imply the identity of the top blockchains and not on their particular value. Ethereum stands out as the most significant blockchain in terms of total value locked (TVL), with the majority of other blockchains maintaining compatibility with its ecosystem. We assume that Polkadot's ability to provide integration between different blockchains enables it to induce its own community. This may signify the future of cross-chain blockchain economic dynamics by introducing cross-chain blockchains that create a more correlated network ecosystem.

Several of Ethereum sidechains (or layer 2 blockchains) such as Polygon and Optimism are not in the same community as Ethereum. This suggests that their properties induce different economic dynamics, causing them to be correlated with other blockchains. The same observation applies to Ethereum Classic and Bitcoin Cash forks, which have separate economic dynamics from their origin blockchains.

Implementing new technologies to connect different blockchains makes the blockchain ecosystem more interconnected, and accordingly, the interoperability graph is becoming increasingly interconnected. However, the blockchain ecosystem seems to have the form of several communities with an internal relatively high correlation of cryptocurrency prices.

V. Data Analysis of Major Blockchains

While in the previous section, we analyzed the complete blockchain ecosystem, in this section we dive into major blockchains and study their interaction. First, we refer to blockchains of high value and high mutual correlation. Next, we further investigate the Ethereum and BNB chains.

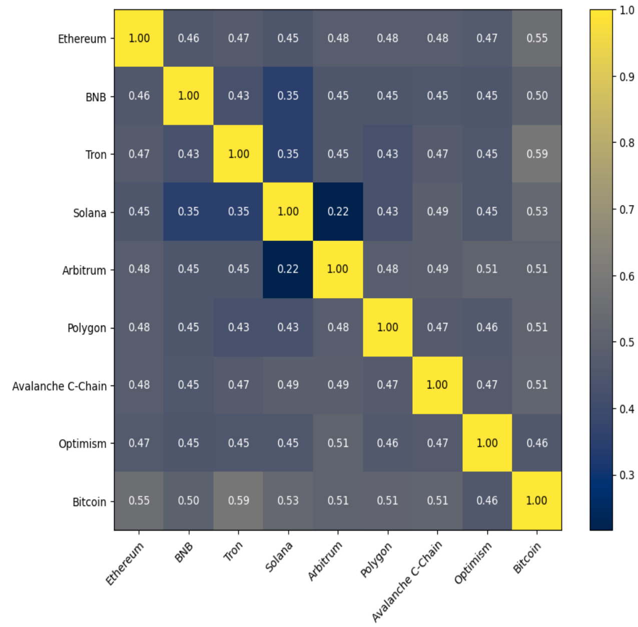


Fig. 7. Correlation between 10 blockchains among the top largest total value locked (TVL), using a threshold of $\tau = 0.4$, from January 21, 2023, to January 17, 2024.

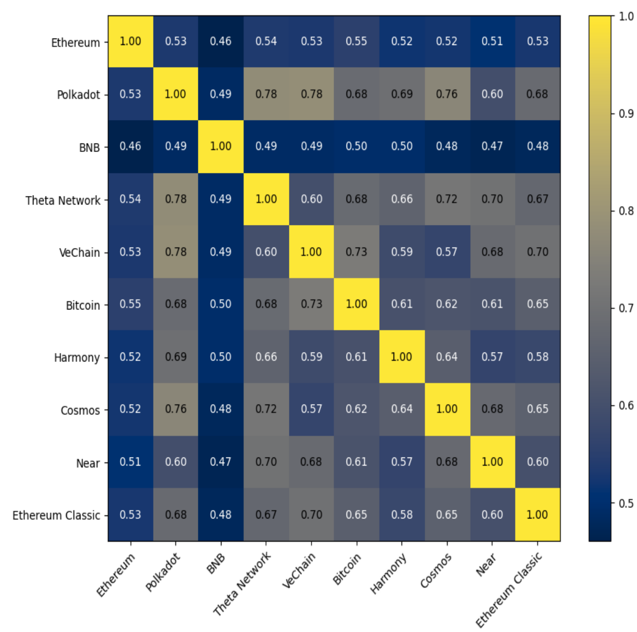
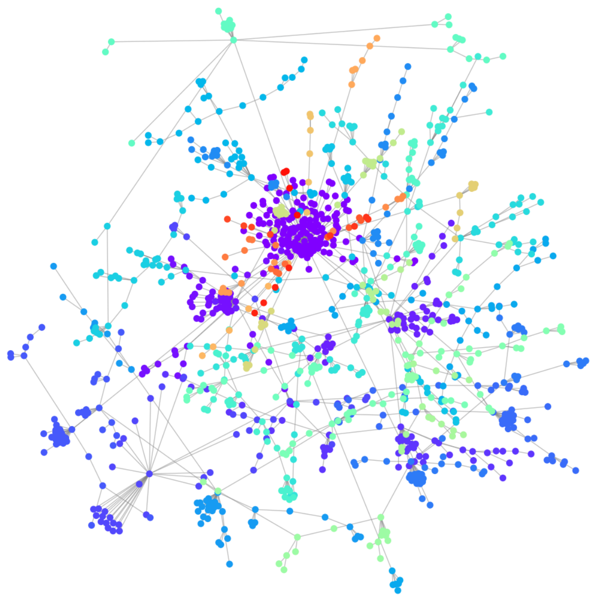


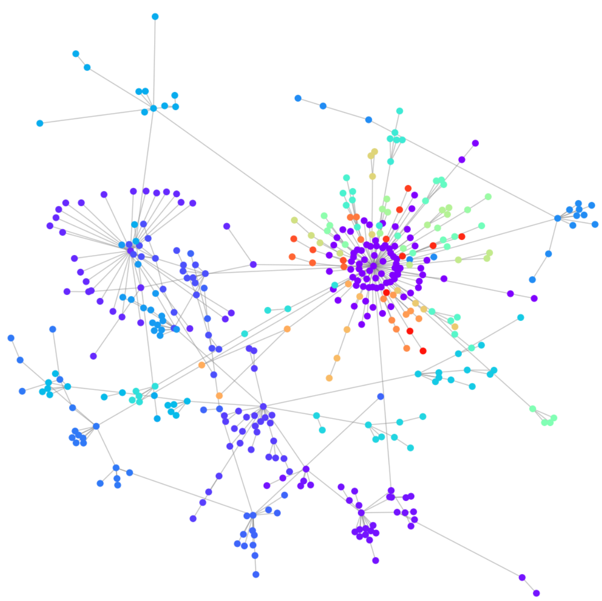
Fig. 8. Correlation between top 10 blockchains with the highest correlation values, using a threshold of $\tau = 0.4$, from January 21, 2023, to January 17, 2024.

A. Correlation of selected blockchains

In this subsection, we examine the correlation between blockchains as discussed in Section III with threshold of 0.4. Instead of creating the Minimum Spanning Tree (MST), we analyze 10 blockchains among the top largest total value locked (TVL) (in Figure 7), and 10 most correlated ones (in Figure 8) in a timeframe from January 21, 2023, to January 17, 2024. We calculate the most correlated blockchains by



(a) Minimum spanning tree of Ethereum using a threshold of $\tau = 0.4$

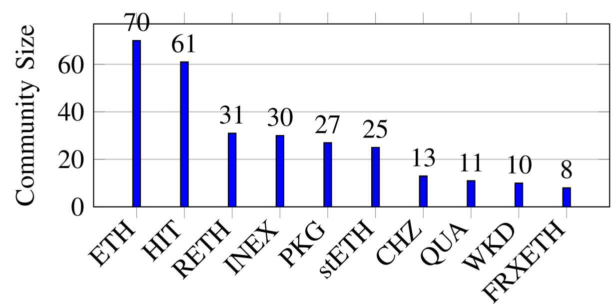


(b) Minimum spanning tree of Ethereum using a threshold of $\tau = 0.6$

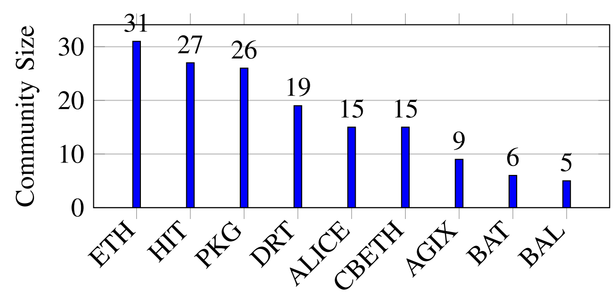
Fig. 9. Minimum spanning tree of Ethereum using thresholds of $\tau = 0.4$ and $\tau = 0.6$, from January 21, 2023, to January 17, 2024. Each color represents a community, with the largest centralized one being around ETH.

summing up their correlations with all others and selecting the top 10. We observe in Figure 7, that the blockchains are moderately correlated with each other, as most correlation values lie between 0.4 and 0.5. This indicates they exhibit similar market dynamics and potentially shared influences in terms of price trends. Further research should be done to explore this behavior. Notably, Bitcoin remains the most correlated among them, showing the highest correlations with each. From Figure 8, we see that Polkadot, Ethereum, BNB, and Bitcoin are among the most correlated, supporting our analysis of the blockchain MST. Additionally, Cosmos, a layer 0 blockchain, appears among the most correlated, which may indicate behavior similar to that of Polkadot. Polkadot displays the highest correlations, closely followed by Bitcoin, which continues to emerge as a top network influence. Moderate correlations are evident in both Figures 7 and 8, indicating that the blockchain ecosystem remains diverse and not dominated by any single blockchain.

B. Ethereum and BNB chain analysis



(a) Ethereum community centers with $\tau = 0.4$

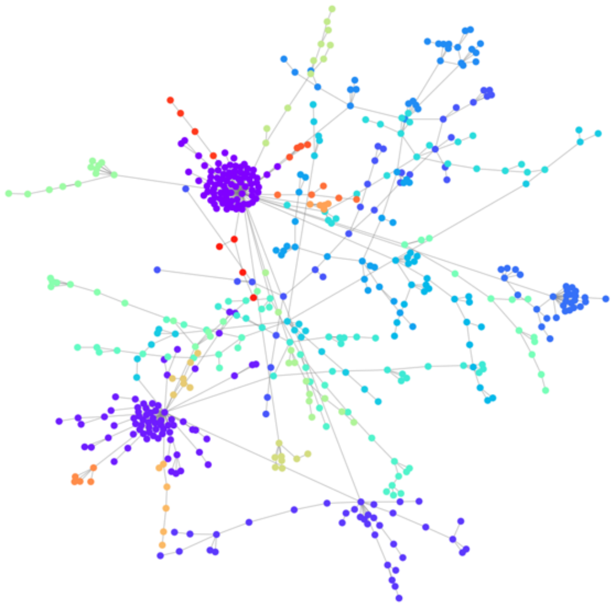


(b) Ethereum community centers with $\tau = 0.6$

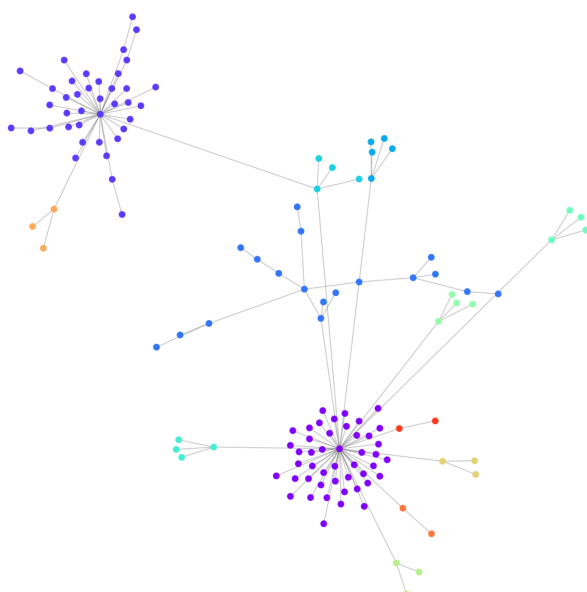
Fig. 10. Comparison of Ethereum community centers at different thresholds of τ from January 21, 2023, to January 17, 2024.

In this subsection we investigate blockchains with high number of cryptocurrencies: Ethereum and BNB to gain insights on the dynamics inside each blockchain. We analyze all cryptocurrencies whose parent blockchains is Ethereum or BNB over the period from January 21, 2023, to January 17, 2024. We use the correlation between these cryptocurrencies with thresholds of 0.4 and 0.6 as described in Section III. We removed edges with a value of 1.0, as this indicates cryptocurrencies that are not correlated. To maintain a connected graph structure, we focus on the largest connected component and consider it as the blockchain itself.

Ethereum Blockchain. From Figures 9(a) and 9(b), we observe that the Ethereum blockchain is not dominated by a single cryptocurrency or a small group setting the price trend. Instead, Ethereum is highly segmented, with 36 communities at a threshold of 0.4 and 16 communities at a threshold of 0.6. Figures 10(a) and 10(b) show that ETH has the largest community, which is reasonable given that it

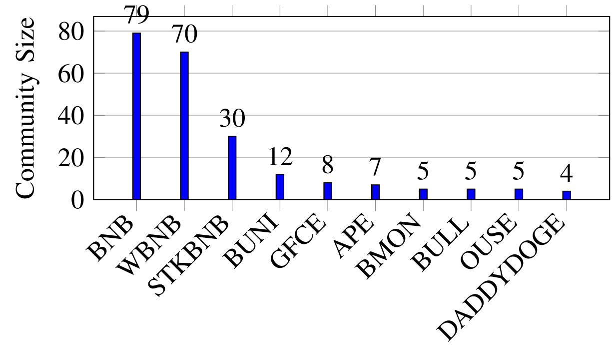


(a) Minimum spanning tree of BNB chain using a threshold of τ = 0.4

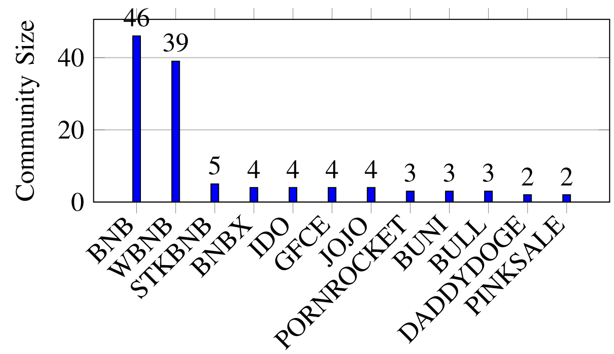


(b) Minimum spanning tree of BNB chain using a threshold of τ = 0.6

Fig. 11. Minimum spanning tree of BNB chain using thresholds of τ = 0.4 and τ = 0.6, from January 21, 2023, to January 17, 2024. Each color represents a community, with the the largest centralized one being around BNB.



(a) BNB chain community centers with τ = 0.4.



(b) BNB chain community centers with τ = 0.6.

Fig. 12. Comparison of BNB chain community centers at different thresholds of τ from January 21, 2023, to January 17, 2024.

is the blockchain's native cryptocurrency. The second-largest community is associated with the HitBTC token, a utility token for the HitBTC exchange. Additionally, PKG Token also ranks among the largest communities; it is a multi-chain, decentralized game and NFT platform token. These tokens represent distinct cryptocurrencies with versatile applications, which, though under-researched, show a considerable influence on price correlations.

BNB Blockchain. From Figure 11(a), we observe that the BNB blockchain appears to contain several centralized communities. The MST is divided into 27 communities, suggesting a degree of decentralization with respect to price trends in BNB. However, as we increase the threshold, as shown in Figure 11(b), we see that the BNB blockchain, with higher-correlated cryptocurrencies, consolidates into two large, centralized communities. This suggests that there is a high number of uncorrelated cryptocurrencies, and as we increase the threshold, we can further investigate the underlying nature of the more correlated cryptocurrencies in the blockchain.

The largest communities in both Figures 12(a) and 12(b) are predominantly composed of cryptocurrencies related to the BNB token, indicating that BNB blockchain may be dominant among more strongly correlated cryptocurrencies. This could support the claims that the BNB blockchain has a more centralized nature [44]–[47].

VI. Investment Implications

Investors can invest in blockchain projects through various means like with capital for application development on a blockchain or by incorporating cryptocurrencies from the same parent blockchain to the portfolio. Our analysis provides potential implications for investment strategies in blockchains.

Market Forecasting. Investors can leverage the correlation between blockchains to gain market insights, which can aid in predicting conditions like a financial crisis or an economic boom. For instance, the correlation of a number of key blockchains can indicate an economic boom or blockchains that become uncorrelated may signal an impending financial crisis. Analysis of the correlation of blockchains can provide information for investors' strategic decisions.

Risk Management and Diversification. Blockchains with high correlations are likely to react similarly to market fluctuations, potentially resulting in positive or negative performance based on overall market conditions. In contrast, uncorrelated ones are likely to react differently. By understanding these correlations, investors can strategically invest in different blockchains based on their correlation to stabilize the portfolio's overall performance. Those same principles are applicable to our analysis of the Ethereum and BNB chains, as we examine their structure in terms of price correlation. Investors can use this analysis to enhance their investment strategies by gaining insights into price trends within each blockchain.

Further work should be done to determine how well our grouping of cryptocurrencies by blockchains and their correlation analysis, impacts portfolios of blockchain investments.

DeFi Lending. Lending is a common DeFi application supported by lending platforms such as AAVE, Compound, MakerDao and Maple [48], [49]. A collateral-based loan allows long lending durations and is typically financed by interest paid by borrower to lenders. Before taking a loan, a borrower provides collateral in one cryptocurrency, allowing it to take a loan in a different cryptocurrency. The loan is over-collateralized: the collateral provided by a borrower has to be worth more than the loan that it covers. To avoid under-collateralization, lending protocols often define minimal liquidation thresholds (with typical values of 120-300%) for various cryptocurrencies. When the collateral-to-loan ratio falls below the threshold, the loan is liquidated and the collateral is sold at a discount. This eliminates the need to return the loan and causes a loss for the borrower. To reduce the chances of liquidation, borrowers can provide collateral of even higher ratios.

Given some minimal liquidation threshold, the initial collateral-to-loan ratio should consider the correlation between the two cryptocurrencies. High correlation is expected to result in often bounded changes in collateral-to-loan ratio. This allows initial values that are close to the threshold, while when the correlation is low, to avoid liquidation, one might consider providing high collateral.

VII. Related Work

To the best of our knowledge, we are the first to study the blockchain ecosystem by the correlation between blockchains. The literature on cryptocurrencies is extensive, given their pivotal role in the blockchain ecosystem.

Dynamics of Cryptocurrency Prices. Cryptocurrency prices have garnered significant attention in academia from various perspectives, such as predicting cryptocurrency prices [50]–[54], analyzing cryptocurrencies' correlation with other financial assets like gold and fiat currencies [55]–[59], and modeling cryptocurrencies volatility [60]–[62]. Our study focuses on the correlation among cryptocurrencies, which is significant for investment decision-making process, portfolio management, risk management, and arbitrage [63]–[67].

Correlation of Cryptocurrency Prices. Several papers have explored the correlation of cryptocurrencies. Such as, Marcin Watorek et al. [68] examine the cryptocurrency market's growth and complexity, comparing it to traditional financial markets. They identify unique characteristics and arbitrage opportunities despite cryptocurrencies' similarities to established markets like the Foreign Exchange Market (Forex). Jiaqi Liang et al. [30] analyze cryptocurrency market risk and dynamics, finding it fragile and unstable despite rapid growth. Correlation matrices and asset trees assess risk, offering investment and regulatory insights. Yongjing Shi et al. [69] apply a multivariate factor stochastic volatility model (MFSVM) to analyze correlations among six major cryptocurrencies, revealing significant positive correlations. They offer insights into the systemic risk of the cryptocurrency market.

Vincenzo Candila [70] utilizes the Dynamic Conditional Correlation model and Google search data to examine interconnections among leading cryptocurrencies. He introduces a novel model, the Double Asymmetric GARCH–MIDAS, highlighting the influence of online searches on cryptocurrency volatility. Boris Radovanov et al. [71] analyze daily prices of Bitcoin, Ether, Ripple, and Litecoin, revealing their volatility and return patterns using autocorrelation and dynamic models like GARCH. Findings indicate persistent volatility with little asymmetry, offering key insights for cryptocurrency investors and portfolio management. Yunus Karaömer [72] explores the impact of policy uncertainty on cryptocurrency returns using the dynamic conditional correlation (DCC) model.

VIII. Conclusions

In this paper, we introduce the concept of the blockchain interoperability graph to measure the connections between blockchains within the ecosystem. Our analysis is based on the correlation between cryptocurrency prices, which are influenced by the interoperability of the parent blockchains they are implemented on. This analysis focuses on specific timeframes using a particular methodology, while other potential evaluation methods and timeframes may yield different conclusions. We highlight several key points from our study:

- Strong correlated cryptocurrencies tend to be more correlated to other cryptocurrencies within their parent blockchains.
- The blockchain ecosystem can be interpreted as a community-based graph, where key blockchains such as Ethereum and Bitcoin are at the center of communities within the graph.
- Blockchains that are forks of their origin blockchains or are sidechains/layer 2 to it tend to form their own economic dynamics.
- Polkadot is an example of a Layer 0 blockchain that fosters a community. Through the analysis, Polkadot remains one of the most highly correlated blockchains, highlighting the potential impact of the protocol's connectivity on the correlation between cryptocurrencies.

- Understanding how blockchains are correlated to each other can offer investment strategies.
- Blockchains with high TVL appear to be correlated with one another, indicating the presence of key blockchains influencing price trends.
- Ethereum and BNB each can be analyzed as an ecosystem with price trends and central cryptocurrencies. In both blockchains, we observe that they are divided in terms of price trends at lower correlation thresholds. However, as the threshold increases, we see the emergence of communities with centralized cryptocurrencies.

Overall, the paper provides insights on the blockchain ecosystem and the interoperability graph based on the correlation between cryptocurrencies.

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Ori Mazor received his B.Sc. and M.Sc. (cum laude) degrees in computer science from the Technion Israel Institute of Technology, Haifa, Israel, in 2021 and 2025, respectively. He is currently a Software Engineer at Microsoft.

Ori Rottenstreich is an associate professor at the department of Computer Science and the department of Electrical and Computer Engineering of the Technion, Haifa, Israel. In 2015-2017 he was a Postdoctoral Research Fellow at Princeton university. Earlier, he received the BSc in Computer Engineering (summa cum laude), and PhD degree from the Technion in 2008 and 2014, respectively.

Appendix A
High-level program design.

We overview the code structure and its main functions as illustrated in Figure 13. **Price Ingestion & Returns.** We collect daily price series for cryptocurrencies from CoinMarketCap using its APIs. We then compute per-cryptocurrency log returns as described in Section III. The output is a returns matrix (cryptocurrencies $\times$ time) that serves as the basis for downstream analyses.

Cryptocurrency-Cryptocurrency Correlation Engine. Using the returns matrix, we compute pairwise correlation coefficients for all cryptocurrencies (Section III). The resulting matrix represents the cryptocurrency correlation graph.

Cryptocurrency-Chain Graph Analyzer. We map each cryptocurrency to its parent blockchain. Using this mapping and several thresholds as described in Section III, we generate the following: Figures 3, 4, and 5.

Chain-Level Correlation Aggregator. We aggregate cryptocurrency-level correlations to the blockchain level by combining, for each pair of blockchains, the correlations of all cross-chain cryptocurrency pairs (Section III). This yields a per-blockchain correlation matrix that represents the blockchain correlation graph.

Inter-Chain Network Constructor. From the chain-level graph, we apply graph sparsification via a minimum spanning tree (and thresholded variants), then compute several network descriptors (degree, betweenness, and community structure). The resulting Figures 6, 7, 8, 14, 15, and 16.

Within-chain cryptocurrency correlations. We analyze the BNB and Ethereum ecosystems, examining the within-chain correlation structure restricted to cryptocurrencies whose parent blockchain is defined for each chain. Following Section III, we build within-chain graphs, construct minimum-spanning trees (MSTs), identify central cryptocurrencies that act as correlation hubs, and compare how these roles evolve across periods. The corresponding subgraphs and hub comparisons appear in Figures 9, 10, 11, and 12.

Appendix B
Illustration of minimum spanning trees

In the following we illustrate minimum spanning trees for three time periods:

(i) January 21, 2023, to January 17, 2024 (a total of 362 days, almost a year) - Figure 14;

(ii) January 21, 2023, to July 18, 2023 (a total of 179 days, roughly six months) - Figure 15;

(iii) July 19, 2023, to January 17, 2024 (a total of 183 days, roughly six months) - Figure 16;

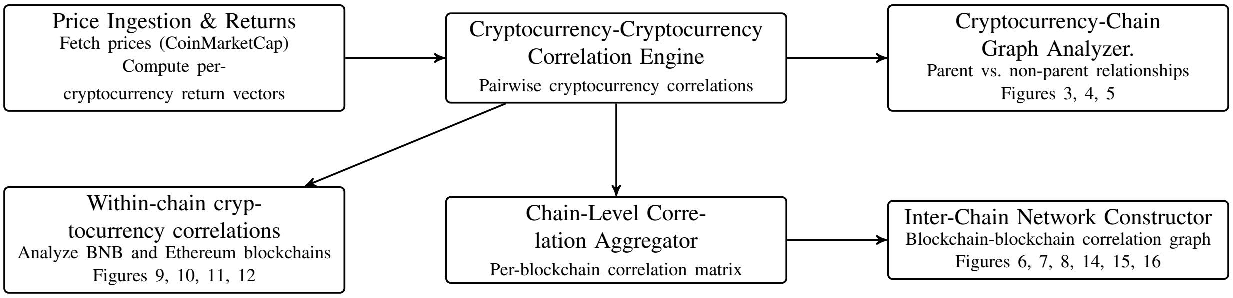

Fig. 13. High level overview of the code structure with its main functions and their dependencies.

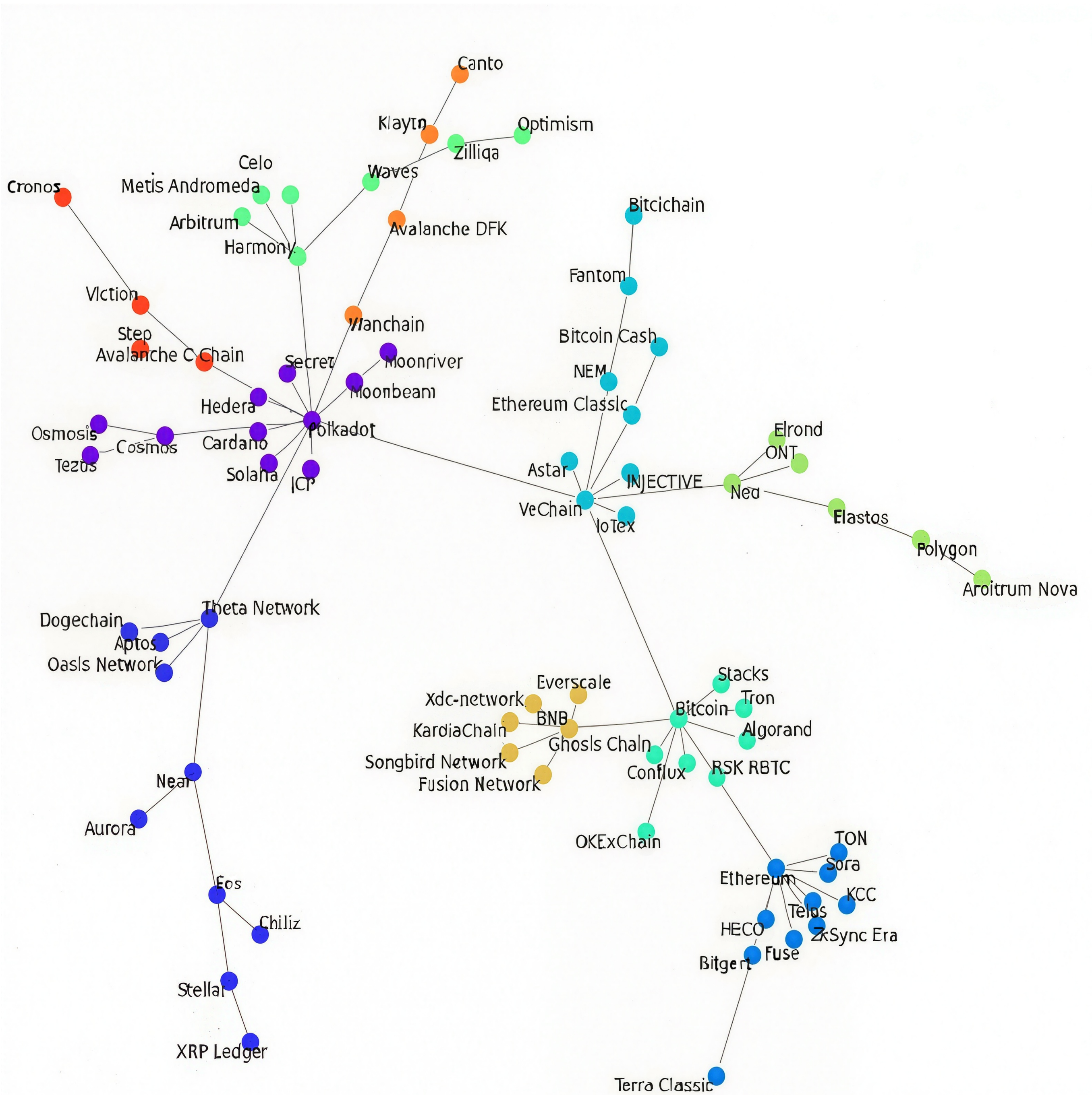

Fig. 14. Minimum spanning tree of blockchains correlation from January 21, 2023, to January 17, 2024.

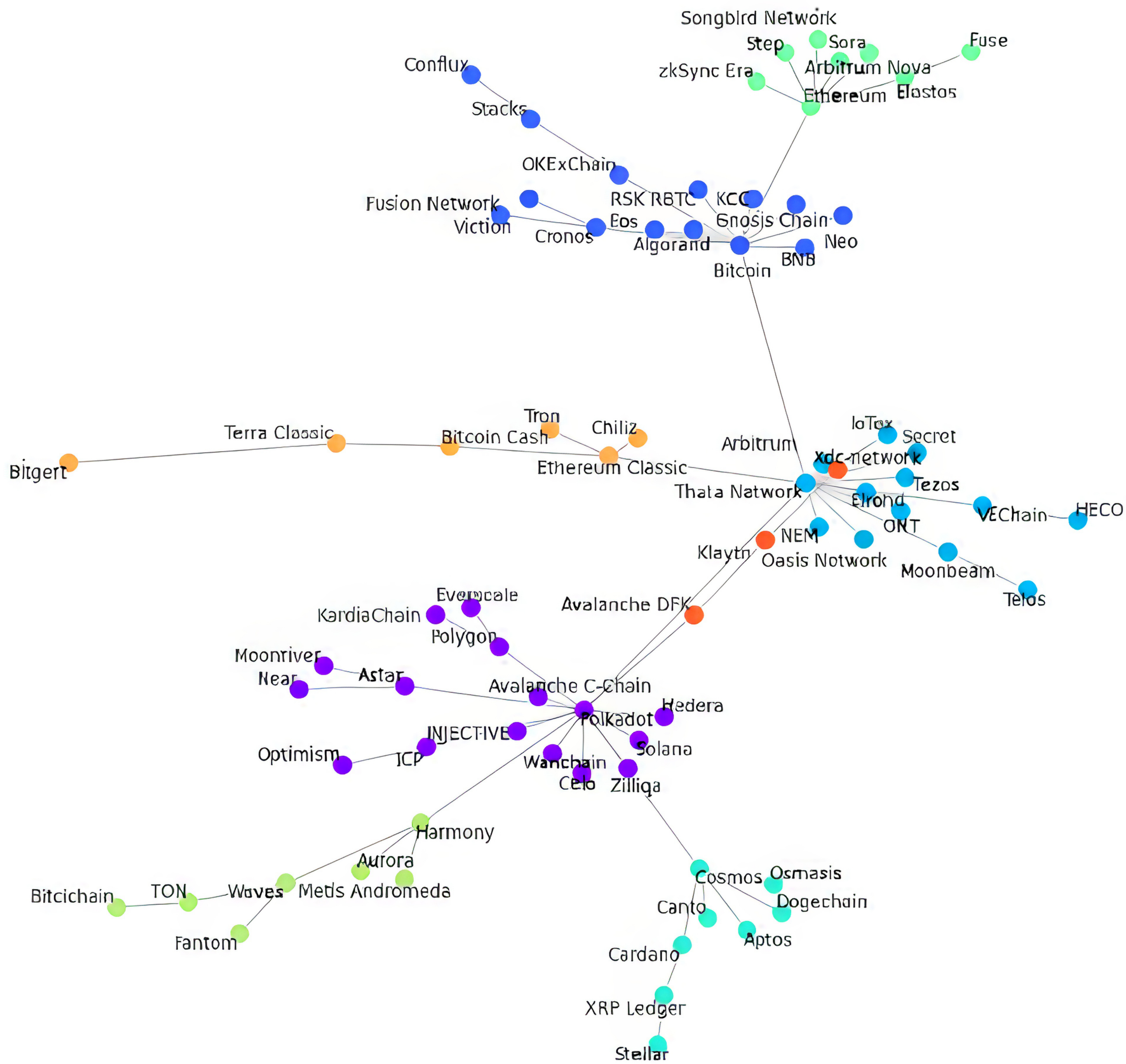

Fig. 15. Minimum spanning tree of blockchains correlation from January 21, 2023, to July 18, 2023.

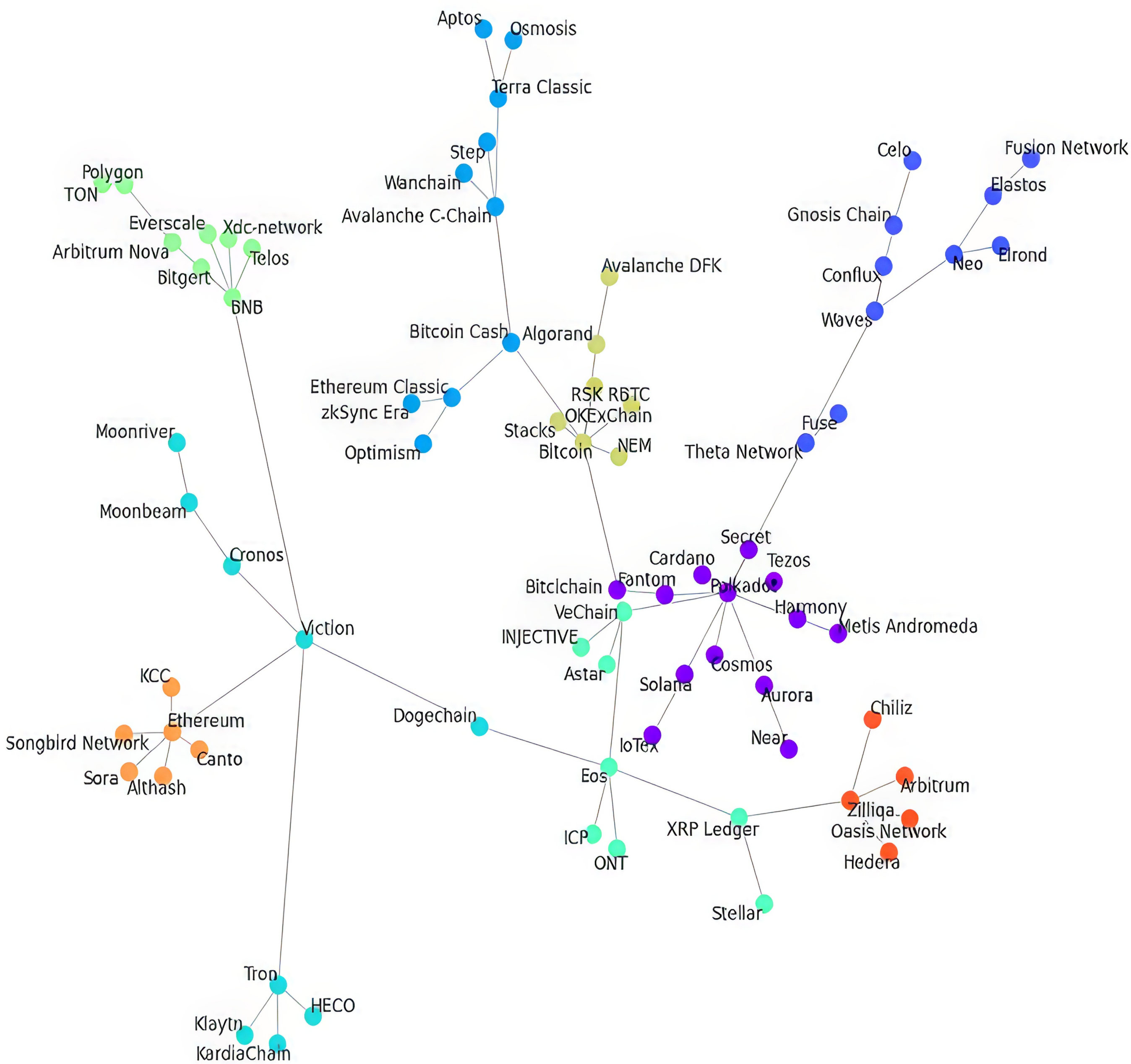

Fig. 16. Minimum spanning tree of blockchains correlation from July 19, 2023, to January 17, 2024.